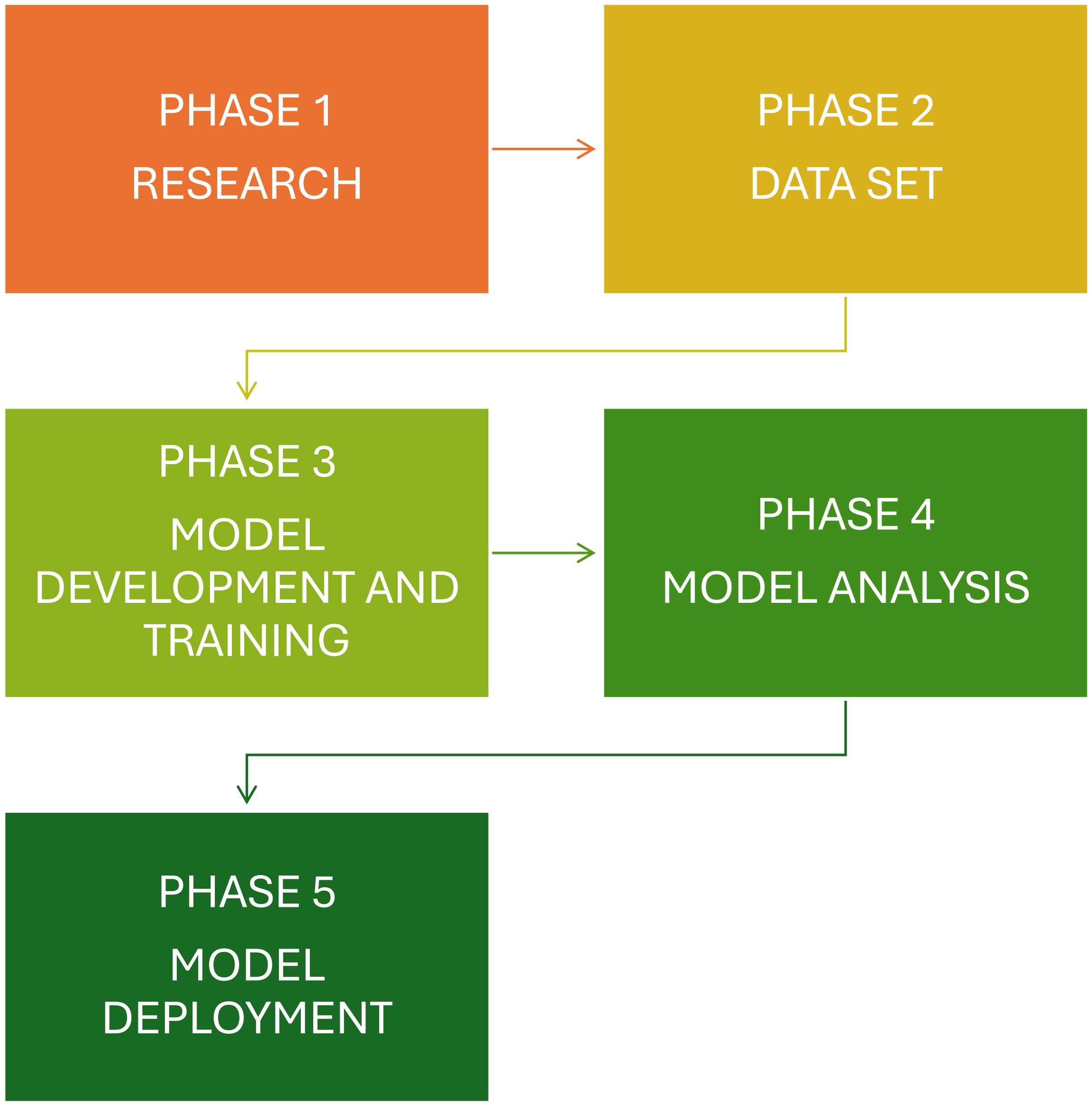
On Device Machine Learning Waste Classification

Presenter: Milton Tinoco

Project Development



Phase 1: Research

Global Challenge:

- Rising pollution levels are damaging ecosystems and human health.
- Recycling is a key method to reduce pollution and conserve resources.

Current Problem:

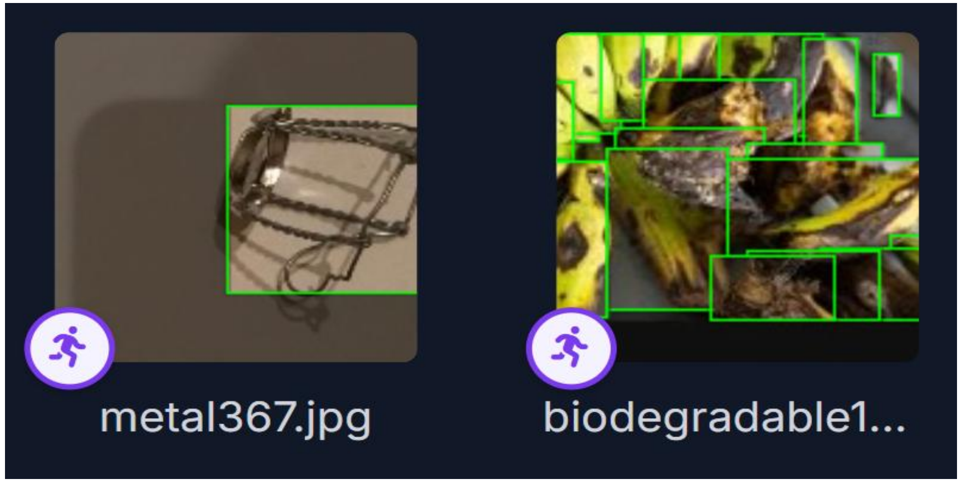
- Confusion about recycling guidelines leads to improper waste sorting.
- Misclassified waste often ends up in landfills, worsening pollution.

Project Solution:

- Developing an on-device machine learning algorithm for waste classification.
- Device will help users accurately sort waste by type (plastic, paper, glass, etc.).
- Supports easy and accessible recycling, reducing environmental impact.

Phase 2: Dataset

Object detection Open-source dataset by **Pumtacho** for waste classification, **Roboflow Universe**.



Category Breakdown 10,000 images:

Biodegradable Objects: **43,381**

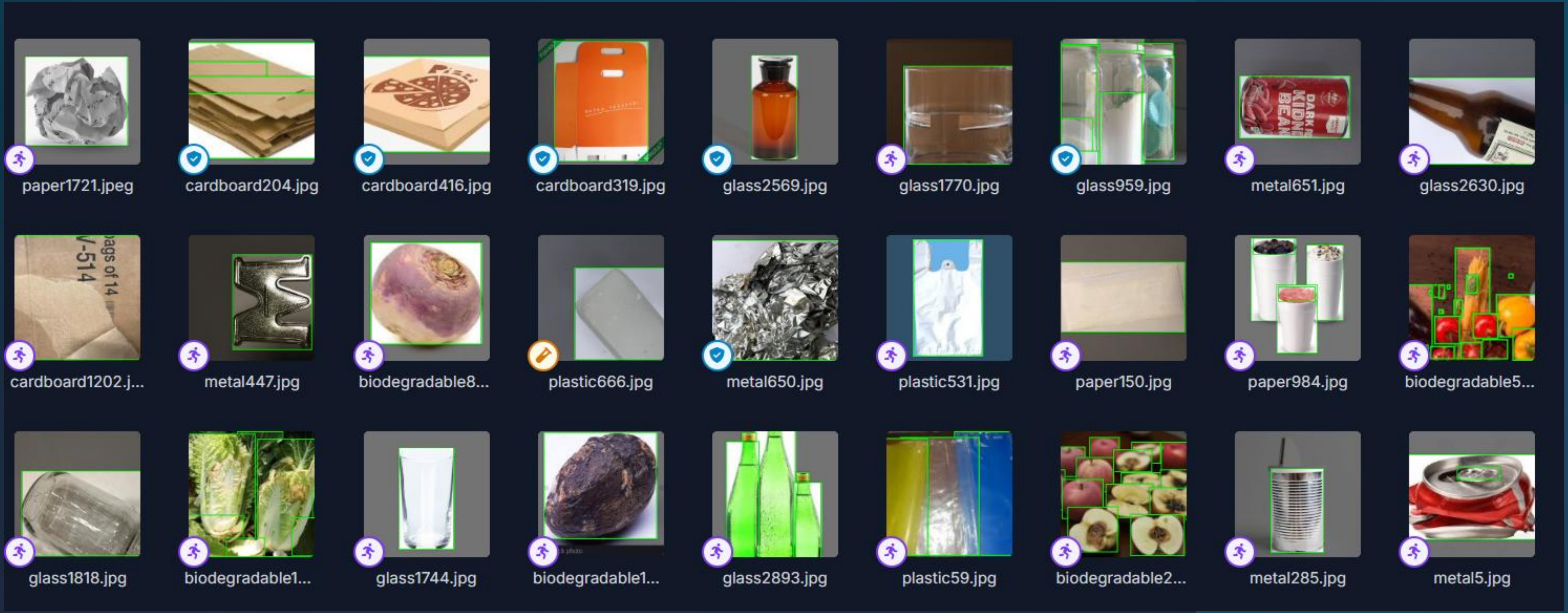
Glass Objects: **7,526**

Plastic Objects: **5,664**

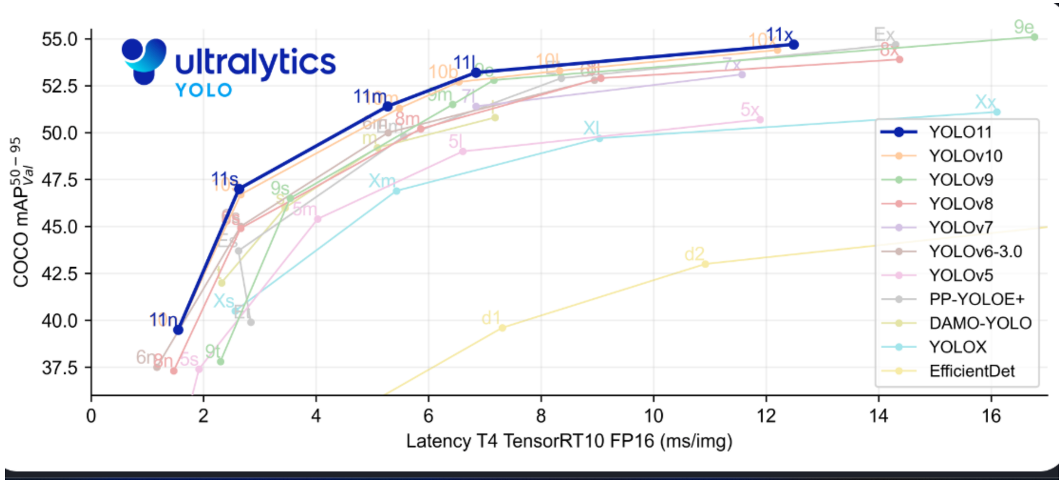
Metal Objects: **5,483**

Cardboard: **4,543**

Paper: **4,279**


paper1721.jpeg
cardboard204.jpg
cardboard416.jpg
cardboard319.jpg
glass2569.jpg
glass1770.jpg
glass959.jpg
metal651.jpg
glass2630.jpg
cardboard1202.j...
metal447.jpg
biodegradable8...
plastic666.jpg
metal650.jpg
plastic531.jpg
paper150.jpg
paper984.jpg
biodegradable5...
glass1818.jpg
biodegradable1...
glass1744.jpg
biodegradable1...
glass2893.jpg
plastic59.jpg
biodegradable2...
metal285.jpg
metal5.jpg

Phase 3: Model Development



Model	size (pixels)	mAP val 50-95	Speed CPU ONNX (ms)	Speed T4 TensorRT10 (ms)	params (M)	FLOPs (B)
YOLO11n	640	39.5	56.1 ± 0.8	1.5 ± 0.0	2.6	6.5
YOLO11s	640	47.0	90.0 ± 1.2	2.5 ± 0.0	9.4	21.5
YOLO11m	640	51.5	183.2 ± 2.0	4.7 ± 0.1	20.1	68.0
YOLO11l	640	53.4	238.6 ± 1.4	6.2 ± 0.1	25.3	86.9
YOLO11x	640	54.7	462.8 ± 6.7	11.3 ± 0.2	56.9	194.9

YOLO11n:

- **Optimized for Edge Devices**: Lightweight and efficient, ideal for Raspberry Pi 4.
- **Fast and Accurate**: Real-time inference with a balanced input size (640px).
- **Pretrained Model**: Supports transfer learning for waste classification.
- **Energy Efficient**: Low power consumption for continuous use.

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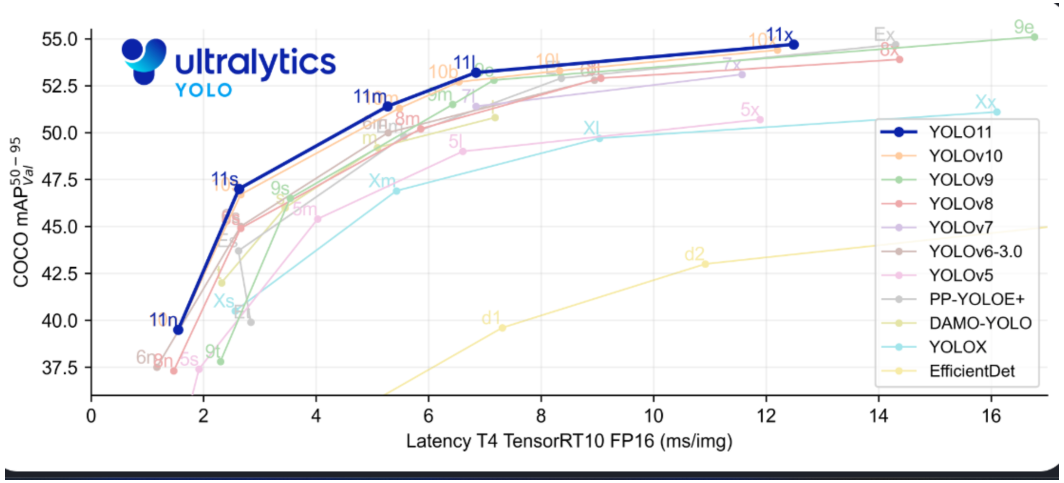

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Phase 3: Training

Model & Dataset:

YOLOv11n.pt with pretrained weights.

Dataset: TrashClass-2 (7002 training, 1997 validation images).

Hardware & Environment:

CUDA (NVIDIA RTX 3060, 6GB VRAM).

Ultralytics 8.3.40, Python 3.12.7, Torch 2.5.1.

Training Configuration:

100 epochs, batch size 16, image size 640px.

Auto-selected optimizer (SGD: lr=0.01, momentum=0.9).

Performance Features:

Automatic Mixed Precision (AMP) for faster training.

Layers partially frozen for transfer learning.

Caching enabled for faster data loading.

Augmentations:

RandAugment, scaling (50%), horizontal flip (50%), HSV adjustments.

```
100 epochs completed in 3.185 hours.
Optimizer stripped from runs\detect\train\weights\last.pt, 5.4MB
Optimizer stripped from runs\detect\train\weights\best.pt, 5.4MB

Validating runs\detect\train\weights\best.pt...
Ultralytics 8.3.40  Python-3.12.7 torch-2.5.1 CUDA:0 (NVIDIA GeForce RTX 3060 Laptop GPU, 6144MiB)
YOLO11n summary (fused): 238 layers, 2,583,322 parameters, 0 gradients, 6.3 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|██████████| 63/63 [00:17<00:00,  3.66it/s]
                   all       1997      18179      0.657      0.531      0.595      0.426
         BIODEGRADABLE        649      13069      0.808      0.522      0.666      0.391
             CARDBOARD        412       1251      0.787      0.528       0.65      0.498
                 GLASS        766       2287      0.921      0.737      0.856      0.669
                 METAL        383       1343      0.846      0.677      0.775      0.554
                 PAPER         15         33     0.0778      0.121     0.0902     0.0732
               PLASTIC         85        196      0.504      0.602      0.534      0.371
Speed: 0.3ms preprocess, 1.7ms inference, 0.0ms loss, 1.5ms postprocess per image
Results saved to runs\detect\train
Ultralytics 8.3.40  Python-3.12.7 torch-2.5.1 CUDA:0 (NVIDIA GeForce RTX 3060 Laptop GPU, 6144MiB)
YOLO11n summary (fused): 238 layers, 2,583,322 parameters, 0 gradients, 6.3 GFLOPs
val: Scanning C:\Users\milto\OneDrive\Desktop\check\TrashClass-2\valid\labels.cache... 1997 images, 0 backgrounds, 0 corrupt: 100%|██████████| 1997/1997
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|██████████| 125/125 [00:19<00:00,  6.31it/s]
                   all       1997      18179      0.657      0.531      0.595      0.426
         BIODEGRADABLE        649      13069      0.811      0.525      0.667      0.392
             CARDBOARD        412       1251       0.79      0.528      0.649      0.496
                 GLASS        766       2287      0.918      0.736      0.856      0.669
                 METAL        383       1343      0.846      0.679      0.775      0.554
                 PAPER         15         33     0.0783      0.121     0.0908     0.0734
               PLASTIC         85        196      0.497      0.597       0.53       0.37
Speed: 0.3ms preprocess, 3.4ms inference, 0.0ms loss, 1.3ms postprocess per image
Results saved to runs\detect\train2
Ultralytics 8.3.40  Python-3.12.7 torch-2.5.1 CPU (AMD Ryzen 9 5900HX with Radeon Graphics)
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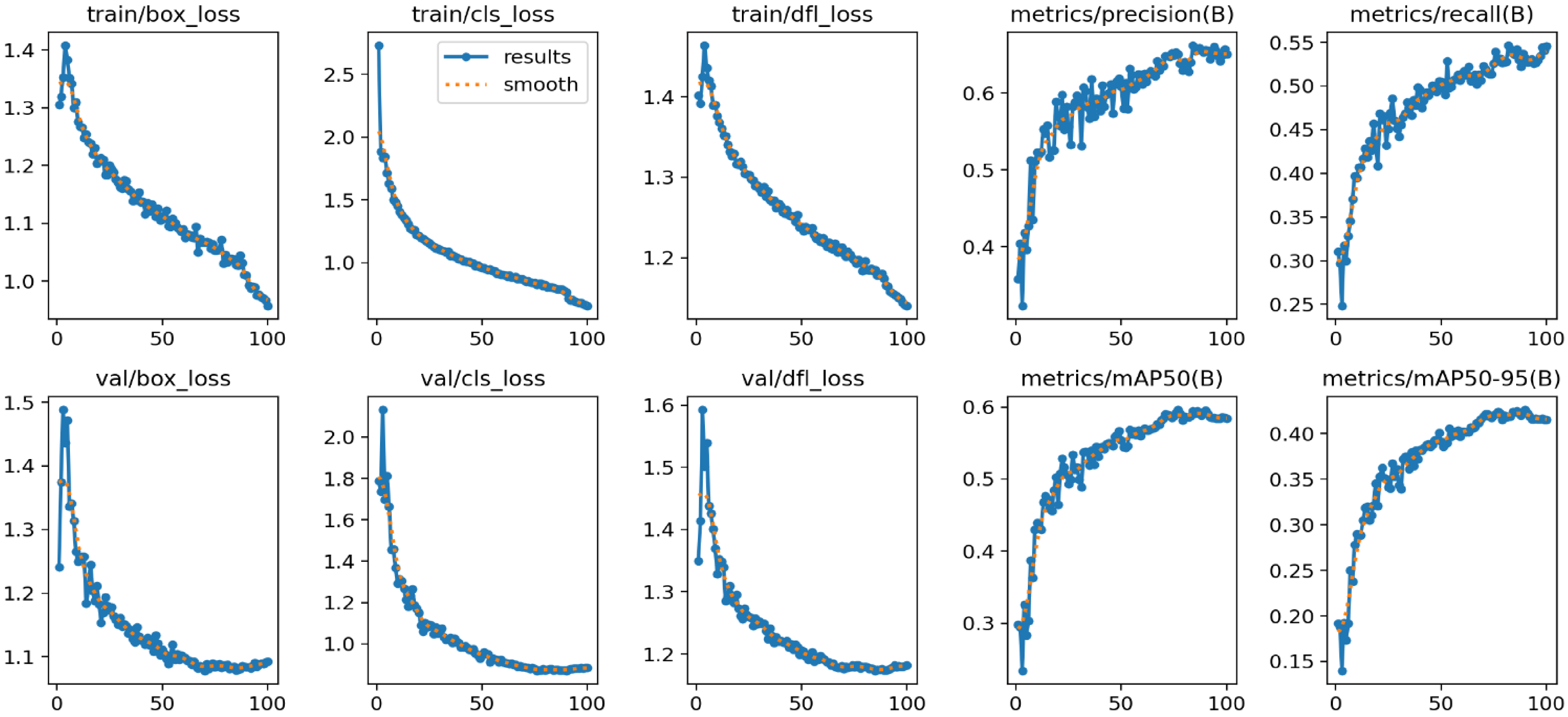
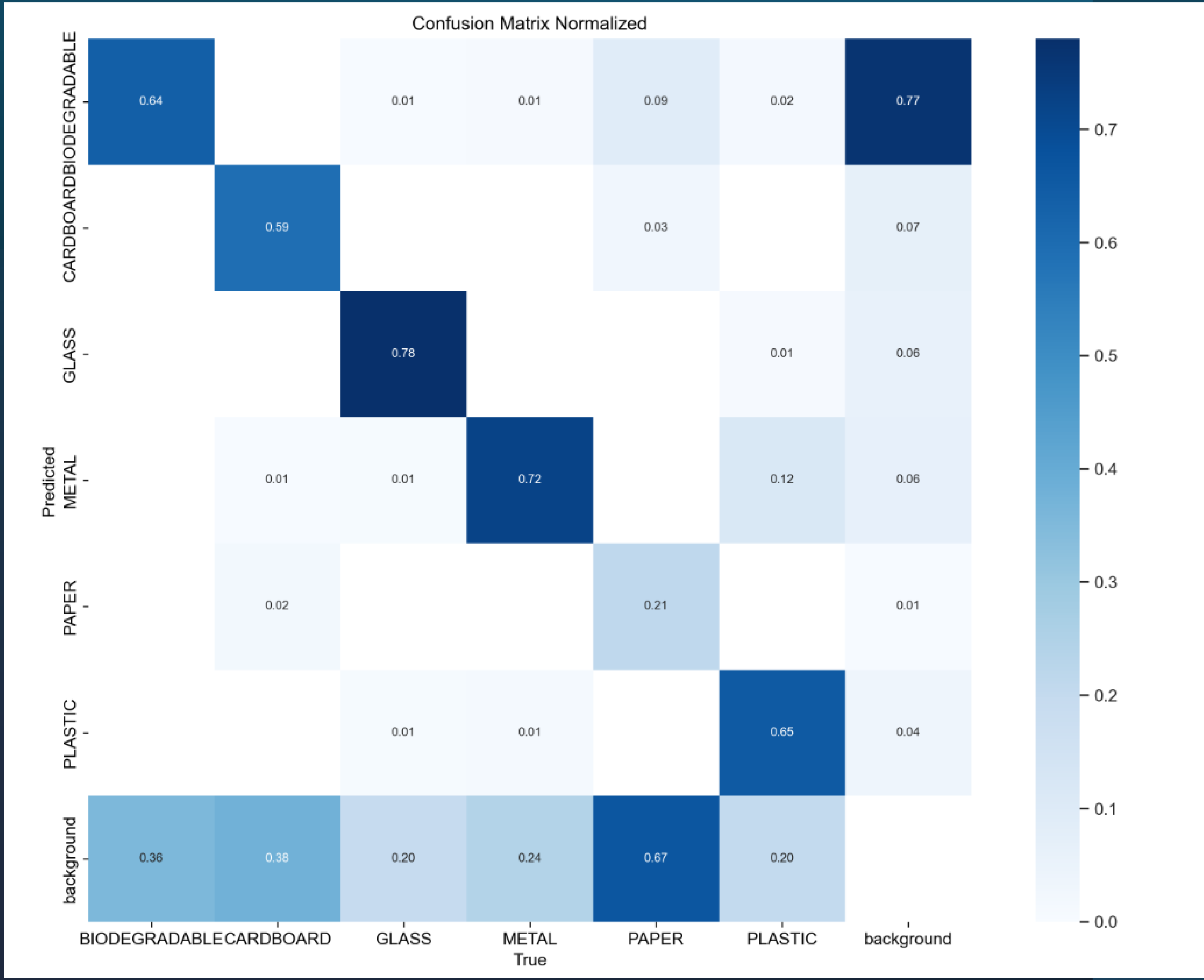
Phase 4: Model Analysis



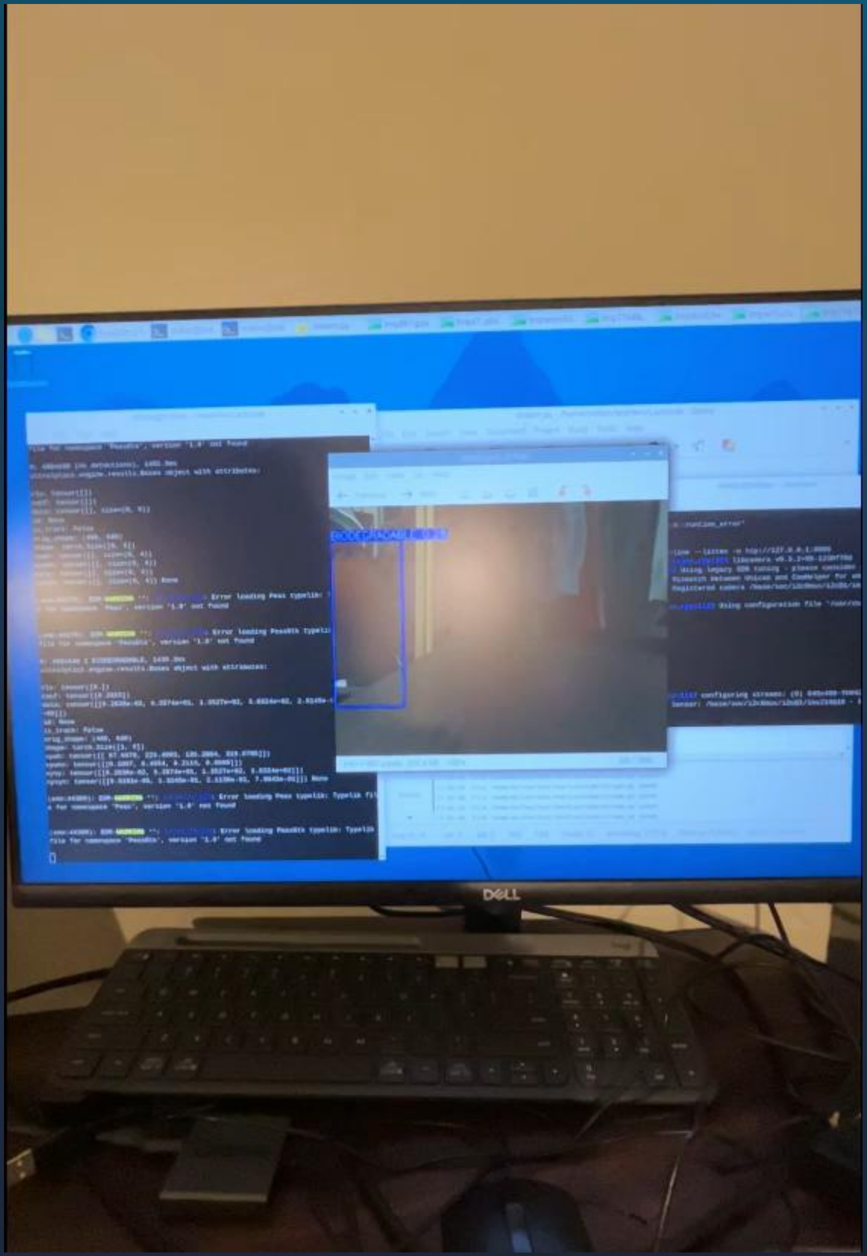

Confusion Matrix Normalized
BIODEGRADABLE
CARDBOARD
GLASS
METAL
PAPER
PLASTIC
background
Predicted
True
0.64
0.01
0.01
0.09
0.02
0.77
0.59
0.03
0.07
0.78
0.01
0.06
0.01
0.01
0.72
0.12
0.06
0.02
0.21
0.01
0.01
0.01
0.65
0.04
0.36
0.38
0.20
0.24
0.67
0.20
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7


BIODEGRADABLE
CARDBOARD
GLASS
METAL
Diclofenac Inj I.P.
www.shutterstock.com • 1765552697

Phase 5: Hardware Deployment:

Raspberry Pi Camera V2

Raspberry Pi 4 Model B 8GB



End Summary

- Successfully selected an optimal dataset and model for waste classification.
- Successfully trained a pretrained YOLOv11 model with 10,000 images for waste classification.
- Achieved balanced performance metrics with an overall precision of 0.657 and recall of 0.531, indicating effective trade-offs between accurate detection and minimizing false negatives.
- The model is lightweight with only 2.6 million parameters and 6.3 GFLOPs, making it ideal for deployment on resource-constrained devices like Raspberry Pi or mobile applications.
- Exceptional performance for detecting glass (mAP@50: 0.856) and metal (mAP@50: 0.775), demonstrating strong detection capabilities for these materials.
- Successfully deployed on Raspberry Pi 4, with readiness for deployment on future devices, including applications.
- Gained hands-on experience in on-device machine learning, with potential expansion for including more waste classification categories in future work.

Future steps

1

Classify Specific Plastics: Enable detection of different plastic types for proper recycling.

2

Improve Challenging Classes: Enhance detection of paper and plastic by increasing dataset size and diversity.

3

Develop an App: Create a user-friendly application for broader accessibility.

4

Expand Dataset: Gather a larger, more diverse dataset to boost performance.

References

- “Americans Are Bad at Recycling. Here’s How the World Does It Better.” *Green America*, www.greenamerica.org/rethinking-recycling/americans-are-really-bad-recycling-only-because-were-not-trying-very-hard. Accessed 27 Oct. 2024.
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- “What Is Waste Segregation: How to Segregate Waste Properly.” *Daily Dump*, www.dailydump.org/pages/learn-waste-segregation?srsltid=AfmBOoq0C53yY_4DM1ds5Jcxyh9L2qRApHjLCYXzIa1FcTg5H12yBNY1. Accessed 27 Oct. 2024.